

Failure Mode & Effect Analysis

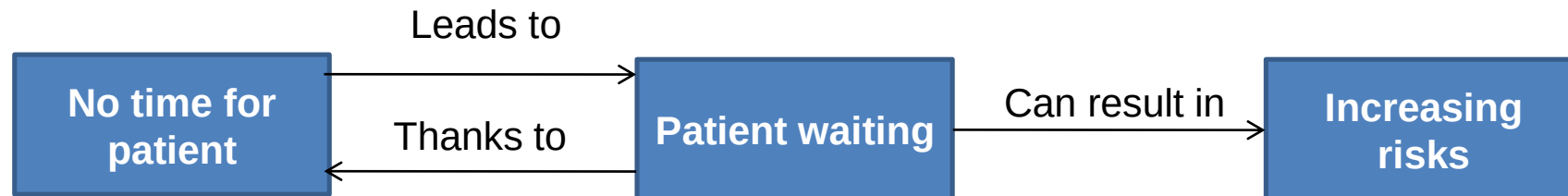
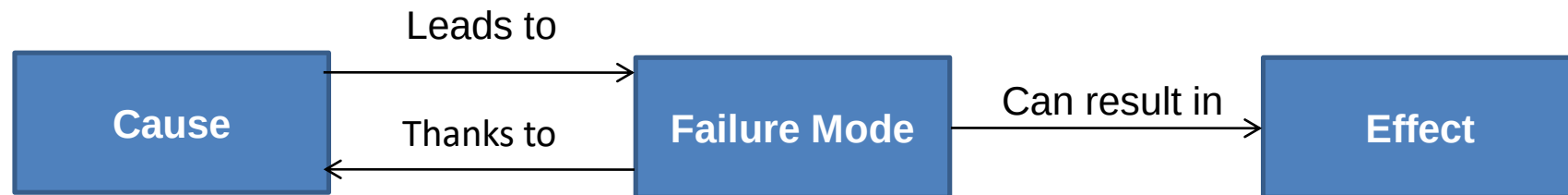
U5 – Analyse

E3 – Analytical Methods

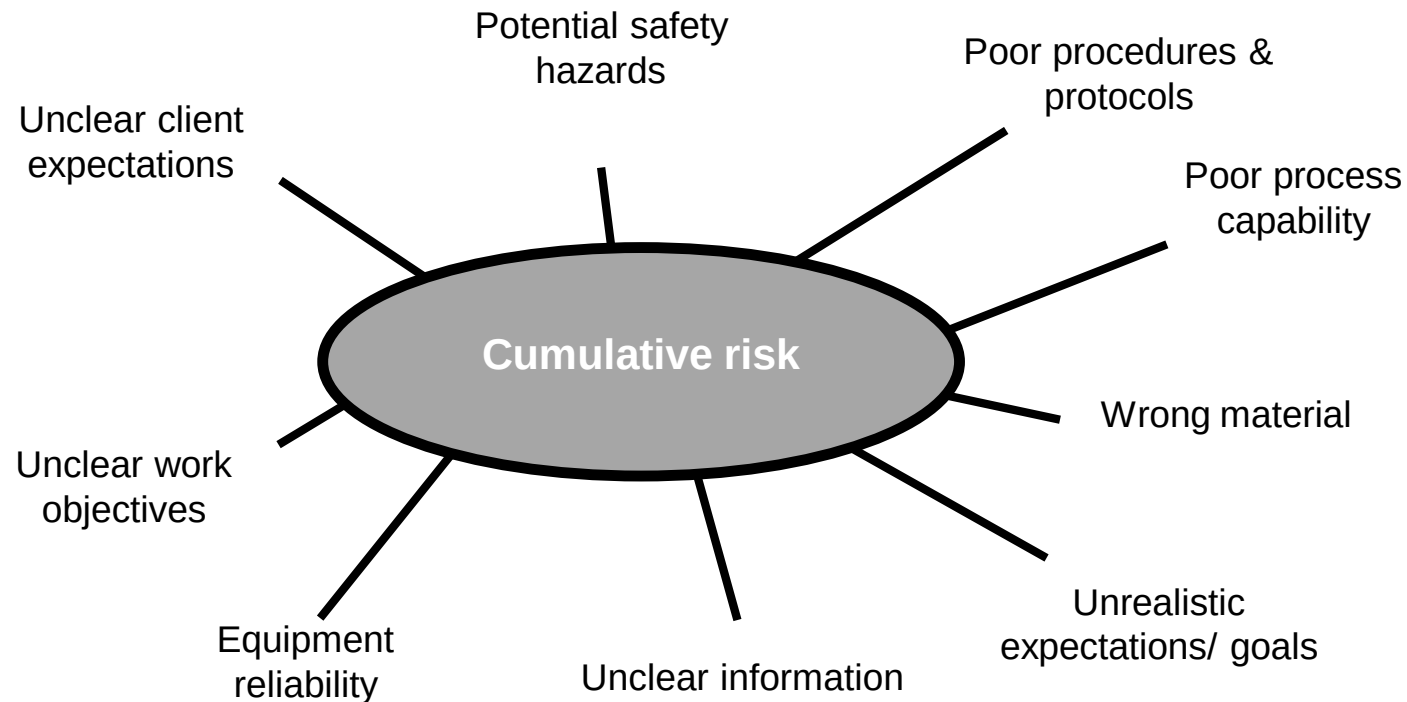
The element 'Analytical Methods' describes the tools that can be used for risk analysis, root cause analysis and waste identification.

Failure Mode & Effect Analysis

Basic principle



Where do risks come from?



D. H. Stamatis, FMEA: FMEA from Theory to Practice, Quality Press, 1995

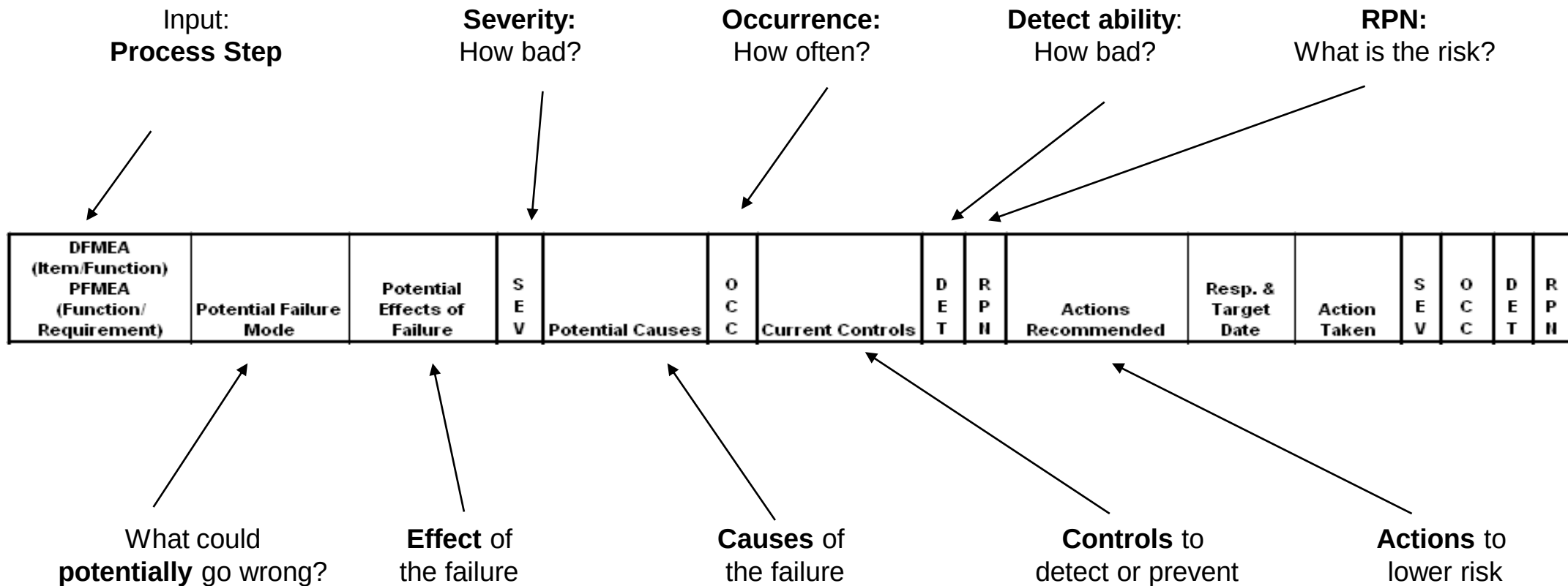
Why is FMEA important?

FMEA is a structured approach to:

- **Identify potential failure modes early and to eliminate their root causes**
- **Reduce the severity of an effect if the root cause can not be eliminated**
- **Reduce the frequency of occurring failure modes**
- **Determine which root causes and failure modes to work on first and which improvement activities must be started first**
- **Increase customer satisfaction by improving the quality, reliability and safety of products**

Failure Mode & Effect Analysis

10 steps to construct a FMEA



Failure Mode & Effect Analysis

Potential Failure Mode

DFMEA (Item/Function) PFMEA (Function/ Requirement)	Potential Failure Mode	Potential Effects of Failure	S E V	Potential Causes	O C C	Current Controls	D E T	R P H	Actions Recommended	Resp. & Target Date	Action Taken	S E V	O C C	D E T	R P H
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Process Step

Failure Mode

Doctor Order of Medication

Wrong drug or dose
Wrong patient

Pharmacy Order Entry

Wrong drug or dose
Wrong patient selected on system

Pharmacy Prep Order

Error in dosage/concentration
Wrong drug selected

Order Administration to Patient

Wrong patient receives drug
Drug administered at wrong time



123RF Stock Photo

Failure Mode & Effect Analysis

Severity

DFMEA (Item/Function) PFMEA (Function/ Requirement)	Potential Failure Mode	Potential Effects of Failure	S E V	Potential Causes	O C C	Current Controls	D E T	R P H	Actions Recommended	Resp. & Target Date	Action Taken	S E V	O C C	D E T	R P H
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- Severity of the effect for the customer or severity of the failure
- Safety is very important

Effect

Severity

Health risk or death	10
Dose is held and patient delayed	3

Rating	Severity of Effect
10	Terminal Injury or death
9	Intervention required, disfigurement
7&8	Major injury permanent lessening of body function
5&6	Minor injury temporary patient harm
3&4	Moderate system problem may affect patient
2	Slight annoyance may affect system
1	No effect

Failure Mode & Effect Analysis

Occurrence

DFMEA (Item/Function) PFMEA (Function/ Requirement)	Potential Failure Mode	Potential Effects of Failure	S E V	Potential Causes	O C C	Current Controls	D E T	R P H	Actions Recommended	Resp. & Target Date	Action Taken	S E V	O C C	D E T	R P H
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- The probability/ frequency that a failure mode/ cause will occur. . . which will lead to the effect
- With the current (preventive) controls
- Without detection measures
- Occurrence refers to how frequently the specified failure is expected to occur
- Occurrence can be estimated on the basis of historical data
For example customer complaints and process know how

Failure Mode & Effect Analysis

Occurrence

DFMEA (Item/Function) PFMEA (Function/ Requirement)	Potential Failure Mode	Potential Effects of Failure	S E V	Potential Causes	O C C	Current Controls	D E T	R P H	Actions Recommended	Resp. & Target Date	Action Taken	S E V	O C C	D E T	R P H
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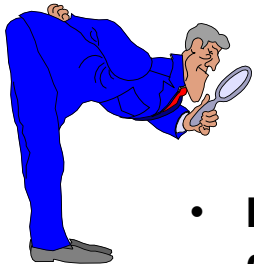
- Can also refer to the frequency of the cases of the failure modes
- 1 = not likely, 10 = very likely

Rating	Likelihood of Occurrence
10	Very high: Failure is almost inevitable
9	
8	High: Repeated failures
7	
6	Moderate: Occasional failures
5	
4	
3	Low: Relatively few failures
2	
1	Remote: Failure is unlikely

Failure Mode & Effect Analysis

Detection

- Possibility of the control to detect the cause or failure mode before the impact can occur



- In other words: what is the probability of detecting the failure or cause?
- 1 = very likely, 10 = not at all

Rating	Ability to Detect
10	Can not detect
9	Very remote chance of detection
8	Remote chance of detection
7	Very low chance of detection
6	Low chance of detection
5	Moderate chance of detection
4	Moderately high chance of detection
3	High chance of detection
2	Very high chance of detection
1	Almost certain detection

Failure Mode & Effect Analysis

Risk priority number (RPN)

DFMEA (Item/Function) PFMEA (Function/ Requirement)	Potential Failure Mode	Potential Effects of Failure	S E V	Potential Causes	O C C	Current Controls	D E T	R P N	Actions Recommended	Resp. & Target Date	Action Taken	S E V	O C C	D E T	R P N
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- The FMEA output is the RPN = Risk priority number
- RPN is based on information provided by:
 - Potential failure modes
 - Effects (impact on client/ patient expectations)
 - Current possibility of the process to detect the failures
- The RPN is calculated as the product of three quantitative ratings, which are related to the effects, causes, and controls:

$$\text{RPN} = \text{Severity} \times \text{Occurrence} \times \text{Detection}$$

Effects

Causes

Controls

Failure Mode & Effect Analysis

Endresult of FMEA

Rating	Severity of Effect	Likelihood of Occurrence	Ability to Detect
10	Terminal Injury or death	Very high: Failure is almost inevitable	Can not detect
9	Intervention required, disfigurement		Very remote chance of detection
8	Major injury permanent lessening of body function	High: Repeated failures	Remote chance of detection
7			Very low chance of detection
6	Minor injury temporary patient harm	Moderate: Occasional failures	Low chance of detection
5			Moderate chance of detection
4	Moderate system problem may affect patient		Moderately high chance of detection
3		Low: Relatively few failures	High chance of detection
2	Slight annoyance may affect system		Very high chance of detection
1	No effect	Remote: Failure is unlikely	Almost certain detection